Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_38

1. Summary of Key Deviations

The analysis of Run_38 reveals a healthy system with no significant faults. All observed deviations are negligible or extremely minor.

- **p_Combustion (Minor):** The only noteworthy deviation is a very slight positive **offset** on the combustion pressure, peaking at around **+4%**.
- **All Other Signals (Nominal):** All other signals across the entire system, including all flows and other pressures, show no significant deviations from their healthy baseline. The vast majority of readings are perfect.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**No Fault**	**Nominal Operation**	All signals are well within their expected operational range. A tiny +4% offset on a single pressure sensor is not indicative of a fault; it represents normal operational variance or an extremely minor sensor calibration drift. There are no critical, major, or physically contradictory deviations that would suggest a problem.
2	**p_Combustion Sensor**	**Very Minor Calibration Error**	The most likely "fault" is simply that the `p_Combustion` sensor has a tiny positive calibration bias, causing it to read slightly high. This would be a sensor-level issue, not a physical problem with the fuel system.

3	**System Running "Hot"**	**Nominal High-End Performance**	The system could be operating at the very high end of its normal performance specifications, leading to a fractionally higher combustion pressure. This is not a fault condition.
4	**Fuel Metering Unit (FMU)**	**Negligible Over-delivery**	An imperceptible over-delivery of fuel from the FMU could cause a tiny increase in combustion pressure. However, this is not supported by any deviation in the `Flow_Engine` signals, making it highly unlikely.
5	**Bypass Valve**	**Negligible High Cracking Pressure**	An imperceptibly higher-than-normal cracking pressure on the bypass valve could lead to a tiny increase in system pressure, but again, the deviations are too small to be considered a fault.

3. Most Probable Hypothesis

The most probable hypothesis is **No Fault**. The system is operating within its nominal parameters.

Reasoning: There are no significant fault signatures in this run. A single, very minor positive offset on one pressure sensor, with no corresponding deviations in any other part of the system, is classic evidence of a healthy system. The observed data falls squarely within the bounds of normal operational variance. There is no problem to diagnose.